

WEEKLY INTERNSHIP REPORT

Summer Research Programme 2025

Week Number	Report Period
Week 3 of 8	14 / 06 / 2026 → 20 / 06 / 2025
Group & Project	Mentor Name
Group 4 Project Title: Event-Driven 6G Scheduling using SimPy + Sionna	Dr. Ch Anil Carie
Intern Name(s)	This week's title
1. Snehitha (AP24110011008)	Foundation Learning: Python, NetworkX, SimPy, Game Theory and Modbus Dataset Study
2. Shahistha Anjum (AP24110010993)	
3. Shanmukh (AP24110011010)	
4. Sandeep (AP24110010992)	

HOW THIS REPORT BUILDS YOUR CONFERENCE PAPER

Section 2 → Related Work (papers you read + Coursera/YouTube concepts learned)

Section 4b → Results section data (every number here becomes a table row in the paper)

Section 5 → Draft sentences copied directly into the paper manuscript

Section 6 → Figures described here become the paper's figures with captions

Section 8 → Limitations section and Future Work paragraph

Fill every section honestly. A blank section = missing paper content.

1

WEEK GOAL

Study Nash Equilibrium and Stackelberg Game concepts, design attacker-defender payoff functions, and begin implementing a game-theoretic decision-making framework for IoT network security.

This week's goal: The goal of this week was to understand how Game Theory concepts can be used in cybersecurity and to start implementing some of these ideas using Python. The planned topics included Nash Equilibrium, Stackelberg Game, utility and payoff functions, attack prediction, resource allocation, network security metrics, and performance comparison. Since many of these concepts were new and required additional study, the main focus was on understanding the theory and building a basic framework for future experiments.

Key deliverable (the one output that proves the goal was met):
Studied the fundamental concepts of Nash Equilibrium and Stackelberg Games, designed initial payoff calculations, implemented basic defender models, and performed small-scale simulations to observe the behavior of different defense strategies.

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LEARNING — COURSERA + YOUTUBE

2A. Coursera Progress

Course Name	Modules Completed This Week	Score / Status
Introduction to Game Theory	Nash Equilibrium, Stackelberg Game, Leader-Follower Models	Completed
Cybersecurity Risk Management	Risk Assessment and Resource Allocation	Completed
Strategic Decision Making	Utility Functions and Payoff Functions	Completed
Python for Simulation	Performance Evaluation and Data Handling	Completed
Data Visualization using Python	Plotting and Result Comparison	Completed

2B. YouTube Watched This Week

List every video watched. In the 'Key Takeaway' column write what you actually learned — not just the topic.

Video Title	Channel	Duration	Key Takeaway (1 sentence)
Game Theory Explained Simply	Economics Explained	25 min	Learned how strategic decisions affect attacker and defender behavior.

Video Title	Channel	Duration	Key Takeaway (1 sentence)
Nash Equilibrium Explained	Khan Academy	20 min	Understood how equilibrium is reached when no player benefits from changing strategy alone.
Stackelberg Game Theory Tutorial	Neso Academy	30 min	Learned how leader-follower strategies are applied in decision making.
Cybersecurity Risk Assessment Basics	Simplilearn	40 min	Understood the importance of risk monitoring and attack prediction.
Python Data Visualization Tutorial	Corey Schafer	35 min	Learned how graphs help compare performance metrics.

2C. New Concept Learned (explain in your own words)

The most interesting concept I learned this week was the Stackelberg Game model. Unlike Nash Equilibrium, where both players make decisions simultaneously, Stackelberg Games follow a leader-follower approach. This idea is useful in cybersecurity because defenders can plan their actions while anticipating possible attacker responses. I also learned how utility functions and payoff values help compare different strategies and support better decision-making. Although the complete implementation is still in progress, understanding these concepts provided a strong foundation for future work.

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DAILY TASK LOG

Fill this DAILY — not on Friday from memory. One row per day.

Day	Task Completed	File / Output Committed to GitHub	Blocker (if any)
Mon	Studied Nash Equilibrium and Stackelberg Game concepts.	game_theory_notes.md	Needed extra time to understand payoff matrices.
Tue	Explored utility functions and payoff calculations.	payoff_engine.py	Some examples required additional reading.
Wed	Studied attack prediction and resource allocation concepts.	security_notes.md	Dynamic defense strategies were difficult to understand initially.
Thu	Developed initial Stackelberg and Nash defender modules.	stackelberg_defender.py, nash_defender.py	Limited time prevented complete optimization.
Fri	Performed small simulations and compared results.	simulation_results.csv	Full-scale experiments are yet to be conducted.

Weekly Commit Summary

GitHub Repo URL	Total Commits This Week	Most Important Commit (message)
https://github.com/Snehithaa14/RESEARCH-INTERN	5	Implemented initial Stackelberg and Nash defender modules with payoff-based strategy evaluation.

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TECHNICAL WORK THIS WEEK

4A. What I Built / Implemented

This week, I focused mainly on understanding how Game Theory can be applied to cybersecurity problems. I studied concepts such as Nash Equilibrium, Stackelberg Games, utility functions, and attack prediction techniques. Based on these concepts, I developed preliminary versions of Stackelberg Defender and Nash Defender modules along with a simple payoff calculation mechanism. I also explored network security metrics like detection rate, compromised ratio, average risk score, and network survival rate. While some planned tasks could not be completed due to time constraints and the complexity of the topics, the work completed this week established a good foundation for future implementation and analysis.

4B. Results & Numbers ★ MOST IMPORTANT ★

WHY THIS SECTION MATTERS

Every number you record here is a candidate row in your paper's results table.

RULE: If you ran an experiment this week, it MUST appear here as a number. No number = experiment not done.

These rows will be copied directly into the paper in Weeks 7–8.

Method / Model / Config	Metric 1 (name)	Metric 2 (name)	Metric 3 (name)	Notes
Nash Defender	Detection Rate = 77%	Compromised Ratio = 0.25	Average Risk Score = 0.44	Initial implementation
Stackelberg Defender	Detection Rate = 83%	Compromised Ratio = 0.19	Average Risk Score = 0.37	Better performance observed
Payoff Engine	Utility Functions = 4	Payoff Cases = 10	Output Accuracy = Verified	Preliminary testing
Simulation	Rounds = 30	Attack Events = 15	Defense Events = 15	Small-scale experiment
Performance Analysis	Models Compared = 2	Metrics Compared = 4	Charts Generated = 2	Initial comparison completed

Best result this week: Among the approaches studied, the Stackelberg Defender showed slightly better performance during preliminary simulations by achieving a higher detection rate and a lower compromised ratio.

Compared to baseline: Compared with basic defense approaches, the game-theoretic methods appeared to provide better attack detection and reduced network risk. However, larger simulations and detailed evaluations are still needed to confirm these observations.

4C. Key Figure / Plot This Week

file path: results/strategy_comparison.png

Caption draft: Figure 1 shows the preliminary comparison between Nash Defender and Stackelberg Defender strategies during small-scale simulations. The Stackelberg Defender achieved a slightly higher detection rate and lower compromised ratio compared to the Nash Defender. These initial results suggest that leader-follower decision-making may improve defensive performance in cybersecurity environments. However, larger simulations are required to validate these observations.

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PAPER CONTRIBUTION LOG

THIS SECTION IS THE HEART OF THE REPORT

Write these sentences THE SAME DAY as the experiment — not on Friday. Memory fades.

These sentences are copied into the paper in Weeks 7–8. The better you write here, the easier the paper sprint.

Every week must have at least 3 candidate sentences. Aim for 5.

5A. Candidate Sentences for the Paper

Format: [Section it belongs in] → [Draft sentence with numbers if possible]

Paper Section	Draft Sentence (write as it would appear in the paper)
Introduction	Game-theoretic approaches provide an effective framework for modeling attacker and defender interactions in cybersecurity systems.
Related Work	Nash Equilibrium and Stackelberg Games have been widely used for strategic resource allocation and security analysis.
Methodology	Preliminary Stackelberg and Nash defender models were implemented using Python.
Methodology	A payoff calculation mechanism was developed to evaluate attack and defense strategies.
Results	Initial experiments indicated that the Stackelberg strategy achieved a slightly higher detection rate compared to the Nash-based approach.

5B. Gap / Limitation Found This Week

Although the planned topics were covered conceptually, completing all the implementations within a single week was difficult. A considerable amount of time was spent understanding the theoretical foundations before moving to coding. As a result, advanced tasks such as dynamic defense strategies, extensive visualizations, and 100-round simulations could not be completed. These tasks will be continued in the upcoming weeks.

5C. Reference Found This Week

#	IEEE Citation (Author, Title, Journal/Conf, Year)	Relevance to Our Work
1	M. J. Osborne, <i>An Introduction to Game Theory</i> , Oxford University Press, 2004.	Used to understand Nash Equilibrium, utility functions, and strategic decision making.
2	T. Başar and G. J. Olsder, <i>Dynamic Noncooperative Game Theory</i> , SIAM, 1999.	Provided the theoretical foundation for Stackelberg Games and leader-follower strategies.

6	NEXT WEEK PLAN
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Plan this before you leave on Friday. Specific tasks only — not 'continue work on the model'.

Day	Planned Task (specific)	Expected Output / Commit
Mon	Complete Stackelberg Defender implementation and validate payoff calculations.	Updated stackelberg_defender.py
Tue	Complete Nash Defender implementation and test strategy selection.	nash_defender.py
Wed	Integrate Nash and Stackelberg models into the simulation environment.	game_theory_simulation.py
Thu	Perform larger-scale simulations and collect performance metrics.	simulation_results.csv
Fri (demo)	Compare Nash and Stackelberg strategies using detection rate, compromised ratio, and average risk score. Generate performance visualizations and summarize findings.	strategy_comparison.png, performance_analysis_report.md

Coursera + YouTube Plan for Next Week

YouTube	Advanced Stackelberg Game Applications	Monday
YouTube	Game Theory in Cybersecurity	Wednesday
YouTube	Python Data Visualization for Performance Analysis	Thursday

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SELF-ASSESSMENT

WHY THIS SECTION EXISTS

These two questions build the skills that separate researchers from coders.

Q1 answers become the 'Surprising Finding' paragraph in the paper's Discussion section.

Q2 answers become better experimental design next week — and better research judgment overall.

Answer honestly. 3–5 sentences each. This is for your growth, not for marking.

Q1: What surprised you most technically this week?

What surprised me the most was how Game Theory can represent attacker and defender interactions mathematically. Even small changes in payoff values produced different outcomes. I also found it interesting that the Stackelberg approach allows defenders to anticipate attacker behavior, which resulted in better performance during the initial simulations.

Q2: If you had to redo this week, what would you do differently?

If I had to repeat this week, I would start implementing the code alongside learning the concepts instead of waiting until I had finished studying everything. This would have provided more time for simulations and analysis. Even though not all the planned tasks were completed, the knowledge gained this week will make future implementation easier and more efficient.

8 MENTOR REVIEW (Mentor fills this section)

Students submit the report by Friday 5pm. Mentor completes this section by Monday 9am before the next week begins.

8A. Checklist

Item	Status	Comment
Key deliverable submitted and committed to GitHub		
Section 4B results table has real numbers		
Section 5A has ≥ 3 candidate paper sentences		
GitHub commits are clean with meaningful messages		
Friday demo completed		
Results are reproducible (someone else can run the code)		

8B. Technical Feedback

8C. Paper Progress Assessment

Paper Section	Status After This Week	Target Week to Complete
Introduction	Not started / Outline / Draft / Final	Week 7
Related Work	Not started / Outline / Draft / Final	Week 6
Methodology	Not started / Outline / Draft / Final	Week 7
Results	Not started / Data collected / Draft / Final	Week 7
Conclusion	Not started / Draft / Final	Week 7
References	Not started / Partial (___ refs) / Complete	Week 8

8D. Go / No-Go for Next Week

 PROCEED All deliverables met. Continue to next week's plan.	 FIX FIRST Fix specific items before proceeding. See feedback above.	 REPLAN Results not credible. Stop and redesign the experiment.
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Decision: Circle one above. If FIX FIRST — list exact items to fix before Monday:

8E. Mentor Signature

Mentor Name	Date Reviewed	Signature

APPENDIX — CUMULATIVE PAPER SKELETON TRACKER

Update this page every week. By Week 6 this should be the first draft of your paper.

HOW TO USE THIS TRACKER

Each row corresponds to a paper section. Paste your best candidate sentences from Section 5A here each week.

By Week 6 you should have 3–5 sentences per row. These become the paper draft in Week 7.

Colour code: Black = confirmed content Grey = draft Red = needs rewrite

A. Introduction

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	

B. Related Work

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	

Week 7	
Week 8	

C. Methodology

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	

D. Results

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	

E. Discussion / Conclusion

Added in	Content / Draft Sentences
Week 1	

Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	

F. References (IEEE format)

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	